

Agile Maturity Self-Assessment Surveys: A Case Study

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Abstract—Agile software development methodologies have been widely adopted by the software industry during the last decade. Agility assessment is an approach to measure the success of this adoption as well as to satisfy the further demands. In response, a number of agile maturity self-assessment surveys have been developed. However, software organizations do not widely utilize existing self-assessment surveys. In this study we aim to identify the existing surveys and evaluate their strengths and weaknesses in agility assessment. Firstly, an exploratory case study is performed by using one of the surveys in order to determine the expected features of an agile maturity self-assessment survey. Then, a comprehensive case study is conducted to measure the sufficiency of the mostly referenced agile maturity self-assessment surveys according to the determined features. The study results reveal that current surveys do not meet the expected features comprehensively.

Keywords— *agility assessment, agility self-assessment, agility surveys, agile maturity self-assessment, agile maturity self-assessment surveys*

I. INTRODUCTION

Agile software development methodologies have been widely adopted by the software industry as a resolution for the problems of the waterfall and process oriented methodologies [1], [2]. Organizations have started to use these methodologies with the purpose of being responsive to change and quick delivery of working products. In order to reach these targets and make the most of the methods, organizations needed to adopt a set of new practices. Most of the time the degree of this adaptation determined the success of the organizations. One approach to measure the adaptation degree is to use the agile maturity assessment. These assessment methods help organizations know where they are and what the improvement points are for improving their agility.

There exist different types of assessment methods for measuring agility of organizations. Some methods are based on traditional assessment approaches which require expert judgement. Others focus on self-assessment methods which aim to assess organizations' agility quickly.

The existing self-assessment methods are usually based on surveys. Some of these surveys were examined in a study

conducted by Jalali et al.[3]. The study focused on descriptive features instead of qualitative properties of a survey. It was found that they have different focuses. The study also includes a case study on two of them which are Comparative Agility [4] and Thoughtworks Agile Assessment survey [5]. It was also found that these surveys give incompatible results as a measure of agility. Gondamoni and Nafchi have also evaluated these two surveys in their study [6]. According to them, these surveys have important drawbacks. They stated that while Comparative Agility includes practices that we are not sure about its suitability for a company, Thoughtworks Agile Assessment survey has insufficient number of practices to measure the agility. In addition to studies related to agility assessment surveys, the tools supporting the application of surveys were also examined in a different study [7]. In that study, tools were examined in terms of the features related to automation.

The existing studies provide limited evaluations about the surveys. They either focus on descriptive features or automation features without examining strengths and weaknesses of the surveys thoroughly by disregarding the content of surveys. In addition, the studies include outdated agile maturity assessment surveys such as Thoughtworks Agile Assessment survey [5] and Nokia Test [8]. All of these show that there is no up-to-date and comprehensive study that examines agility assessment surveys in detail.

The purpose of our study presented in this paper is to identify the strengths and weaknesses of the surveys in agility assessment. To achieve this goal we performed an exploratory case study by using one of the agility assessment surveys in order to determine the expected features of an agility assessment survey. Then, we conducted a case study in one medium-sized media company with the mostly referenced agility assessment surveys to evaluate sufficiency of the surveys in terms of the expected features.

Following the introduction, this paper is organized as follows: In section 2, we explain exploratory case study. In section 3, we detail the evaluation criteria obtained from exploratory case study. In section 4, we describe the case study design and conduct. In section 5, we present findings obtained

during the case study in detailed way. Finally, in section 6, we provide conclusion and future work.

II. EXPLORATORY CASE STUDY

The aim of the exploratory case study is to determine evaluation criteria in order to measure the capability of existing agility assessment surveys. We aim to find expected features of a survey. For exploratory case study, we determined following two research questions.

RQ 1: What are the strengths and weaknesses of the survey?
RQ 2: What are the expected features of an agile maturity self-assessment survey?

In the following subsections, we explained how we designed and conducted the exploratory case study, and then presented findings.

A. Design of the Exploratory Case Study

In the design of our exploratory case study, we adopted the following strategies:

Survey Selection Strategy: We planned to select the survey which has the highest number of reference in academic publications in order to have a better idea about a commonly referenced survey.

Case Selection Strategy: Our strategy was to select an organization which we had performed agility assessment based on AgilityMod [9] reference model. The reason for this selection was to perform the assessment quickly since we were able to identify weaknesses and strengths of the organization.

Data Collection Strategy: The assessor would take notes about the usage of selected survey by considering advantages and disadvantages of the survey while performing self-assessment with the survey.

Data Analysis Strategy: In order to answer the research questions, we decided to use content analysis strategy. For research question 1, we would collect the findings in categories which were weaknesses and strengths. For research question 2, we would use all items in both categories in order to determine the features that an agile self-assessment survey must have.

Validation Strategy: After performing assessment, we planned to present the results to each assessor who had performed the assessment in the organization.

B. Conduct of the Exploratory Case Study

From limited number of academic publication, Comparative Agility [4] is the most common agility assessment survey. It is cited in 7 academic papers obtained at the end of literature review in well-known databases: IEEE, ScienceDirect, Web of Science and SpringerLink. Therefore, we decided to use Comparative Agility in assessing the agility of the organization to be selected.

Comparative Agility was developed by Mike Cohn and Kenny Rubin in order to compare organizations in terms of their agility according to 8 areas. Its aim is to give the agility position of an organization by comparing with its competitors.

We selected an agility assessment performed on IT organization that the assessors assessed its agility. We performed the agility assessment in an organization which is a solution provider for information and communication technologies in local and global market. It is located in Ankara, Turkey. A team developing an information system project was examined in terms of agility. The project was started one year ago and 600KLOC has been developed with PHP Language. The project team consists of 4 software developers, 1 test engineer, 1 product manager and 3 part-time business analysts.

The assessment was performed by one of the assessors who took active role in the previous assessment. After completing assessment, the results were reviewed by another assessor who performed the previous assessment based on AgilityMod reference model [10]. The person who performs the assessment took notes about the advantages and disadvantages of Comparative Agility.

Then, final findings to answer each research question was reviewed by an assessor who has at least ten years' experience on process improvement and two years' experience on agile software development methodologies.

C. Findings of the Exploratory Case Study

We discuss the findings in two sub-sections: findings related to agility of the organization and findings related to Comparative Agility.

1) Findings Related to Agility of the organization

Table 1 Agility Assessment Results

Area	Teamwork	Requirements	Planning	Technical Practices	Quality	Culture	Knowledge-Creating	Outcomes
Question								
1	+0.86	+0.11	-1.83	+0.82	+0.15	+0.02	+0.06	-2.68
2	+0.94	+1.40	+0.94	-1.01	+1.39	+0.29	-0.18	-2.77
3	+0.60	+0.22	+0.83	-1.49	+1.11	-1.44	+0.19	+1.25
4	+0.72	-0.13	-0.05	+0.36	-1.83	-0.81		+1.27
5	-0.16	-1.62	-1.64	+0.50	+0.20	+0.82		+1.14
6	+0.98	+0.97	+1.08	+0.89	+1.06			+1.24
7	+1.15	+1.00	+0.54	+0.33	+0.89			
8	-0.91		+0.29	-2.15				
9	+0.51		+0.27	-1.54				
10	-0.08		+0.31					
11	+0.34		+0.19					
12	-1.09		-2.27					
13	-0.04		+0.91					
14	+0.76							
Average	+0.30	+0.28	-0.02	-0.34	+0.37	-0.25	+0.02	+0.08

The results obtained after assessment with Comparative Agility are shown in terms of standard deviation of item and area. They were automatically deduced from the analysis of 18757 assessment results in database of the survey. From 8 areas, 5 of them are above average while 3 of them are below average.

2) Findings Related to Comparative Agility

Comparative Agility is a five-point likert type survey. That is, each question can be answered by choosing one option from five options or one option “Not Applicable”. The assessment shows the remaining questions in term of percentage and the remaining time over 15 minutes in the survey.

Assessment results show the standard deviation for each dimension and each question based on the answer of all organizations in the database. The deficient points which have negative standard deviation can be seen from the results. That is, an organization can identify improvement opportunities in a certain area in comparison to a benchmark set. However, there is no improvement suggestion about how to enhance their implementation in the organization.

The results are shown with the number of standard deviation and explain what they mean. Moreover, there are scoring values in terms of standard deviation for each dimension and question. However, there is no general scoring value or agility level obtained with analysis of the assessment results for the organization.

The questions are short enough to read and answer. However, there are some questions that can be considered as subjective. For example, the answer for the question “The team is more productive than before” may be different according to team member who performs the assessment.

The survey can be answered based on all types of agile methods since it focuses on generic agile methods rather than one specific agile method. It is suitable to be used by a single team member and does not allow multiple users to perform self-assessment. All of these findings obtained during assessment are listed as strengths and weaknesses of the survey. These findings answer our first research question in the case study.

The strengths of the survey are listed below:

- Inclusion of eight areas consisting of diverse agile practices
- Performing assessment easily with 5 point likert scale
- Observability of improvement areas from assessment results
- Good explanation of scoring
- Scoring of each area of organization
- Asking one thing at a time
- Having short questions
- Being suitable for different agile software development methodologies

The weaknesses of the survey are listed below:

- Not having improvement suggestions
- Not indicating general agility of organization
- Having subjective questions which cannot be measured

- Not allowing multiple team members to perform the assessment

These items related with strengths and weaknesses of the survey are categorized by getting inspiration from the criteria in a study [11] about the comparison of agility assessment models. These categorization answers our second research question in the exploratory case study. At the end of analysis, following 7 features are determined for an agile maturity self-assessment survey to have.

Table 2 Expected Features of an Agile Maturity Self-Assessment Survey

Feature	Item
Comprehensiveness	- Inclusion of eight areas consisting of many practices
Fitness for purpose	-Performing assessment easily with 5 point likert type -Observing improvement areas from assessment results -Not having improvement suggestions
Discriminativeness	-Having meaning and explanation of scoring -Scoring of each area of organization -Not indicating general agility of organization
Objectivity	-Having some subjective questions which cannot be measured
Conciseness	-Asking one thing at a time -Having short questions
Generalizability	-Being suitable for each kind of agile software development methodology
Suitability for multiple assessment	-Not allowing multiple team members to perform assessments

III. EVALUATION CRITERIA

In this section, we detailed each feature that we had obtained at the end of the exploratory case study.

A. Comprehensiveness

A self-assessment survey must include all agile practices. Therefore, we check whether a survey includes the practices in AgilityMod since it is the newest and the most comprehensive agility reference model [9]. Other than AgilityMod, there are some agility reference models [12]–[16]. However, according to Leppanen, other models are in beginner level [17]. In addition, other models do not meet completely the expected features such as scope, definition of agile levels, objectivity and completeness [11]. AgilityMod has been developed to be coherent with agile values and principles, based on the meta-model of ISO/IEC 15504 [18] process assessment model. It includes 34 agile practices that an agile organization has to have. Therefore, each question or item in a survey is matched with the most suitable practice in AgilityMod. We rated this criterion based on surveys’ coverage of these 34 agile practices using following rating scale:

- Not Achieved: 0-5 agile practices are covered.
- Partially Achieved: 6-17 agile practices are covered.
- Largely Achieved: 18-29 agile practices are covered.
- Fully Achieved: 30-34 agile practices are covered.

B. Fitness for Purpose

A self-assessment survey must be developed with the purpose of assessing team's or organization's agility and assisting organizations in software process improvement. Therefore, assessment must include questions or items that can be evaluated or checked. In addition, results must include deficiencies and improvement suggestions that guide the organizations on how to become more agile. Therefore, we consider following 3 criteria while performing the case study.

- 1- Questions or items can be marked with a tick or a circle.
- 2- Assessment results include deficiencies in detailed way.
- 3- Assessment results include detailed improvement suggestions.

We shaped following rating scale according to these 3 sub-criteria:

- Not Achieved: None of them is provided.
- Partially Achieved: Only one of them is provided.
- Largely Achieved: Two of them are provided.
- Fully Achieved: All of them are provided.

C. Discriminateness

A team or an organization must be able to make distinction between poor and good agility. Therefore, they must be able to determine their levels of agility and assign a descriptive label or number to their agility. Therefore, an agile maturity self-assessment survey must meet following 3 criteria.

- 1- Assessment results include the meaning and explanation of scoring.
- 2- Assessment results include agility level of organization.
- 3- Assessment results include scoring of each area of organization.

We shaped following rating scale according to these 3 sub-criteria:

- Not Achieved: None of them is provided.
- Partially Achieved: Only one of them is provided.
- Largely Achieved: Two of them are provided.
- Fully Achieved: All of them are provided.

D. Objectivity

At the end of a self-assessment, verifiable results must be produced. The judgment of the assessor must be at a minimum level. Therefore, each question must carry same meaning for everyone and must be measurable. We evaluated each question one by one to determine whether related question is measurable or not. Then, we determined the objectivity for each practice using four-point scale according to distribution of measurable questions in each practice. Finally, we rated a self-assessment survey based on ratings of practices using following rating scale:

- Not Achieved: 0%-15% of practices are rated as "Fully" or "Largely".

- Partially Achieved: 16%-50% of practices are rated as "Fully" or "Largely".
- Largely Achieved: 51%-85% of practices are rated as "Fully" or "Largely".
- Fully Achieved: 85%-100% of practices are rated as "Fully" or "Largely".

E. Conciseness

Each question in a survey must be short and ask only one thing. Therefore, we evaluated each question one by one to determine whether related question is clear or not. The questions that ask more than one thing or are long to read are determined as "not conciseness". Then, we determined the conciseness for each practice using four-point scale according to distribution of concise questions in each practice. Finally, we rated a self-assessment survey based on ratings of practices using the same rating scale in determining objectivity.

F. Generalizability

A self-assessment survey must be usable for all kinds of agile methods. Therefore, it must be developed with the purpose of assessing generic agile methods, not only one or two specific agile methods. The rating for generalizability is performed according to following rating scale:

- Not Achieved: No agile method is addressed
- Partially Achieved: Only one agile method is addressed.
- Largely Achieved: More than one agile method are addressed.
- Fully Achieved: Generic agile methods are addressed.

G. Suitability for Multiple Assessment

In order to get more accurate results, assessment must be performed by different team members. Therefore, a self-assessment survey must be developed to be suitable with multiple assessments. In order to prevent subjectivity, a self-assessment survey must support multiple assessments by enabling team members fill surveys individually and analyzing results based on assessment of all team members.

- Not Achieved: Only single team member can fill survey
- Partially Achieved: More than one team member can fill survey and there is no analysis.
- Largely Achieved: More than one team member can fill survey and it includes overall analysis.
- Fully Achieved: More than one team member can fill survey and it includes in depth analysis.

IV. THE CASE STUDY

In this section, we explained how we designed and conducted the case study in order to answer following research questions.

RQ 1: To what extent do surveys cover the features that are identified in the exploratory study?

RQ 1.1: Do the surveys meet agile practices in AgilityMod?

RQ 1.2: Are the surveys fit for purpose?

RQ 1.3: Are the surveys discriminative enough to determine the agility?

RQ 1.4: Do the surveys have objective questions?

RQ 1.5: Do the surveys have concise questions?

RQ 1.6: Can the survey be used for all kinds of agile methods?

RQ 1.7: Are the surveys suitable for multiple assessment?

A. Design of the Case Study

In the design of our case study, we adopted the following strategies:

Survey Selection Strategy: We planned to make a literature review in order to determine agile maturity self-assessment surveys which are published as scientific papers. We also used search engines to determine commercial agility assessment surveys. The criterion for selection from those identified surveys is the availability and popularity of agile maturity self-assessment surveys.

Case Selection Strategy: Our strategy was to select an organization having assessment results about their agility assessment. Therefore, we planned to perform assessment on an organization that assessors had already assessed their agility. The reason of choosing such an organization was that we were smoothly able to reach evidences about weaknesses and strengths of the organization related to its agility. Therefore, an organization which had already assessed by professional assessors according to AgilityMod reference model would be selected. In addition, we planned to select a different organization from the organization in our previous exploratory case study.

Data Collection Strategy: In order to collect data, we decided to use a spreadsheet consisting of aspects and practices of AgilityMod, a form consisting of 7 expected features and assessment reports. The assessor having experience on agility assessment based on AgilityMod reference model was expected to match each question or practices with the practices in AgilityMod and fill the form during the assessment. At the end of the assessment, the assessment report is obtained about the agility of the organization. We also planned to examine this assessment report in terms of expected features.

Data Analysis Strategy: Our plan was firstly to write number of questions for each practice after finishing the assessment. Then, we aimed to calculate number of covered practices according to existence or absence of a question for each practice in order to determine comprehensiveness.

In order to find the objectivity and conciseness, we concentrated to examine each question or item one by one in terms of objectivity and conciseness. Then, we would

determine the objectivity and conciseness of the survey according to number of “Largely Achieved”, and “Fully Achieved” questions.

In order to find sufficiency of other features, we decided to develop a case description for each case. We would evaluate the case description and form including seven features with content analysis. That is, we would match the findings with the criteria, and then rate each criterion for a survey according to 4 point scale mentioned in evaluation criteria section.

Validation Strategy: After performing assessment with each survey, we planned to prepare assessment reports and discuss the results with an expert of both process improvement and agile software development methodologies.

B. Conduct of the Case Study

According to literature review, 8 available self-assessment surveys were determined as the mostly referenced surveys. The following table shows the most popular surveys:

Table 3 List of Agile Maturity Self-Assessment Surveys

Name	Survey Owner	Type	Number of Questions/Items
42-Point Test [19]	Kelly Waters	Yes/No	42
Agile Karlskrona Test [20]	Mark Seuffert	Multiple Choice	11
Agility Questionnaire [21]	Marcel Britsch	6 point likert scale	60
Borland Agile Assessment [22]	Borland	7 point likert scale	12
Comparative Agility [4]	Mike Cohn and Kenny Rubin	5 point likert scale	65
Depth of Kanban [23]	Christophe Achouiantz	5 point likert scale	69
Scrum Master Checklist [24]	Michael James	5 point likert scale	42
Scrum Self Assessment [25]	Cape Project Management	Yes/No	60

After that, we decided to perform the case study with an available organization that professional assessors had already performed assessment so that we could reach assessment results. From the organizations, we selected the organization which we have the most knowledge and resources about. As a result, we owned detailed assessment results with in-depth analysis about the agility of an organization.

The organization is one of the leading media companies in Turkey with its 17 million unique visitors on its various internet platforms. An ongoing online video platform project including 9 software developers, 2 graphical user interface designers, 2 business intelligence analysts, 1 tester and 8 content providers was assessed according to AgilityMod reference model. The evidences were collected from two project managers, a software team leader and a graphical user interface designer via interviews. The assessment results based

on AgilityMod reference model showed that exploration and construction aspects of the organization were in first levels while its transition and management aspects were not implemented.

During this study, agility assessment was performed again with each available agility assessment survey in the light of assessment results and the evidences collected from organization. Since we reached 8 agility assessment surveys, we performed 8 different assessments with the same organization in different times.

As mentioned in data collection part, we took notes in a form according to seven evaluation criteria and examined each question to match them with the practices in AgilityMod. After finishing the assessment, we obtained an assessment result based on related survey if the survey supports to generate assessment result.

We analyzed the forms, the spreadsheets, the assessment results, and case descriptions as mentioned in data analysis part in case study design section. Then, we rated each agility assessment surveys using the same rating approach [26] defined in ISO/IEC 15504 in terms of each evaluation criteria based on the definitions mentioned in evaluation criteria section.

After rating assessment, the results were presented to an information expert who has at least 10 years' experience on process assessment and 2 years' experience on agile software development.

C. Validity Treats

The survey-based assessments were performed by one of the authors of this paper. He has assessment experience based on AgilityMod [9], CMMI [27] and ISO/IEC 15504 [18] since 2014. However, he had not taken active role on the previous assessment process of the selected case based on AgilityMod reference model. Therefore, there was a possibility to assess agility of the case incorrectly. In order to avoid this possibility, he examined the expert-based case study results and evidences in detailed way before starting case study with each agile maturity survey. Since he has enough experience on agility assessment process, it was easy to understand and adopt the expert-based case study results. In addition, after performing survey-based assessments, the results were discussed with an expert of both process improvement and agile software development methodologies in order to eliminate any bias of the assessor. This increased the internal validity. Furthermore, external validity is very important in order to generalize the results. Therefore, sampling carries vital importance to represent all agile maturity self-assessment surveys. Thus, we selected agile maturity assessment surveys as many as possible since our aim is to evaluate the capability of the surveys. In terms of construct validity which refers to the degree of measuring what is expected to measure, we selected an organization having enough number of indicators for agility assessment so that we can increase the degree of measuring capability of the surveys.

V. RESULTS

The following table shows the assessment results obtained at the end of the case study for each agile maturity self-assessment survey according to determined features.

Table 4 Case Study Results

Survey / Criteria	Comprehensiveness	Fitness for Purpose	Discriminativeness	Objectivity	Conciseness	Generalizability	Suitability for Multiple Assessment
42-Point Test [19]	PA	PA	PA	FA	LA	FA	LA
Agile Karlskrona Test [20]	PA	PA	LA	FA	FA	FA	NA
Agility Questionnaire [21]	PA	PA	LA	LA	LA	FA	NA
Borland [22]	PA	PA	NA	PA	LA	FA	NA
Comparative Agility [4]	LA	LA	LA	LA	FA	FA	NA
Depth of Kanban [23]	PA	LA	FA	FA	FA	PA	NA
Scrum Master Checklist [24]	PA	PA	NA	LA	PA	PA	NA
Scrum Self Assessment [25]	LA	PA	LA	FA	FA	PA	NA

Comprehensiveness: Assessment results show that there is no fully achieved survey in terms of comprehensiveness. In other words, they do not focus on all agility aspects sufficiently. Many of them concentrate on mostly management aspect. In addition, the surveys generally focus on specific aspect attributes such as iterative and learning whereas they disregard the attributes related with simplicity and technically excellence. There are only 2 surveys meeting largely the practices in AgilityMod. These surveys are Comparative Agility and Scrum Self Assessment. While Comparative Agility includes likert type questions, Scrum Self Assessment includes true/false questions. The common properties of both surveys are that they include sufficient number of questions for more practices when compared to others.

Fitness for Purpose: Even though the surveys claim that they are developed with the aim of assessing agility of a team or an organization, there is no survey which meets fully the feature "fitness for purpose". They do not generate assessment results that include the missing points and improvements suggestions. The purpose of the assessment is to guide the organization on continuous improvement as well as identifying problems. Almost all of the surveys focus to identify problems only. Therefore, many of the surveys are partly suitable for assessment purpose. There are only 2 surveys which largely achieves fitness for purpose. These are Comparative Agility, and Depth of Kanban. Both surveys do not include any improvement suggestion to increase the agility level of the organization. The common positive property of these two surveys is to support reporting of assessment.

Discriminateness: Assessment results show that only one of the surveys provides discriminateness fully. This survey is Depth of Kanban. In the survey, there are defined agility levels that show the agility degree of an organization based on the specific scoring range. It includes four agility levels. These levels are “Necessary for Sustainable Improvements”, “Improving Sustainably”, “Excellence” and “Lean”. All of these are indicated with colors in radar chart. In addition, the survey results include areas or features that give more specific results about the agility of the organization. In Depth of Kanban, these are related to 7 properties of Kanban: Visualize Effects, Improve, Feedback Loops, Explicit Policies, Manage Flow and Limit WIP. There are also some practices to increase the agility in the survey. Apart from this survey, half of the surveys are largely discriminative. Comparative Agility, Agile Karlskrona Test, Scrum Self Assessment, and Agility Questionnaire are largely achieved surveys in terms of discriminateness. They do not include either agility level for the organization or scoring for areas/aspects of agility. The remaining are either partially achieved or not achieved. The general reason of not meeting discriminateness fully is the missing comprehensive assessment methods in the surveys.

Objectivity: Half of the surveys include measurable questions that lead to objectivity: 42-Point Test, Agile Karlskrona Test, Depth of Kanban, and Scrum Self Assessment. The scaling type of most of these surveys is true-false. They are generally either checklists or marked as true or false. Only Agile Karlskrona Test has multiple-choice questions. From other surveys, Comparative Agility, Scrum Master Checklist, and Agility Questionnaire are largely achieved in terms of objectivity. They include questions related to especially productivity and efficiency such as “the team is more productive” or “the team produces higher quality products”. Only Scrum Master Checklist is partially objective. It includes the words leading to subjective answers such as “good”, “better”, and “well”.

Conciseness: There are 4 surveys that meet conciseness fully. These surveys are Comparative Agility, Agile Karlskrona Test, Depth of Kanban, and Scrum Self Assessment. They are asking one-question at a time. In addition, the lengths of the questions are short enough to answer. While 42-Point Test, Borland, and Agility Questionnaire are largely achieved in terms of conciseness, Scrum Master Checklist is partially achieved. These surveys include questions that connect two different things. An example from 42-Point Test is the item “Software is tested and working at the end of each sprint/iteration.” Testing software and delivering working software are two different things that need to be considered separately. General property of them is that two different concepts are connected to each other with a conjunction such as and, but etc.

Generalizability: Most of the surveys focus on generic methods rather than focusing on one specific method. From the surveys, three of them focus on only one agile software development method. These surveys are Depth of Kanban, Scrum Master Checklist and Scrum Self Assessment. That is, they are not generalizable to all agile software development methodologies. 2 of them which are Scrum Master Checklist, and Scrum Self Assessment are based on Scrum while Depth

of Kanban is based on Kanban. In other words, these surveys are developed to be compatible with only one specific agile software development method. They are partially applicable to other methods since all agile software development methodologies share same principles. Apart from these three surveys, it is suitable to use other surveys in each kind of agile methodologies to assess organization’s or team’s agility.

Suitability for Multiple Assessment: None of the surveys meets expectations fully for multiple assessment. Only one of the surveys is largely suitable for multiple assessment. The largely achieved survey is 42-Point Test. Even if the survey is text-based, it can be used for multiple assessment. In the explanation of 42-Point Test, it can be seen that it allows each team member of an agile team to fill the survey. In addition, it gives superficial information about the analysis. However, it is not possible to get detailed analysis from the survey. Other surveys do not support multiple assessment. They are only suitable for single agility assessment. In other words, no more than one team member can perform self-assessment with these surveys.

VI. CONCLUSION

In this study, 8 widely available, most frequently referenced agile maturity assessment surveys are examined by means of a case study in terms of seven criteria determined at the end of an exploratory case study. These criteria are Comprehensiveness, Fitness for Purpose, Discriminateness, Objectivity, Conciseness, Generalizability, and Suitability for Multiple Assessment. The case study results show that none of the agile maturity self-assessment surveys has fully satisfied the expected features. Especially, there is no survey meeting the expectations fully in terms of comprehensiveness, fitness for purpose and suitability for multiple assessment. There is only one survey which is suitable largely for multiple assessment. In addition, two surveys are largely comprehensive whereas two surveys are largely fit for the purpose. Only one of them (Comparative Agility) meets largely both features. Furthermore, the most disregarded feature is the suitability for multiple assessment where seven of the eight surveys do not have this feature while the most motivated feature is generalizability where five surveys had fully achieved scores.

From the 8 surveys, Comparative Agility had 6 features which are largely or fully achieved. However, it also has significant improvement opportunities. It meets only two features which are conciseness and generalizability completely. The survey is largely comprehensive, fit for the purpose, discriminative, and objective. It requires increasing number of covered practices, a way to suggest practices for improvement, and having agility level definitions and more measurable questions. In addition, it is only suitable for single assessment. Therefore, it will be more helpful if it is used by an expert assessor.

Other than Comparative Agility, many surveys focus on 4 features even if these features are not fully achieved. These surveys are 42-Point Test, Agile Karlskrona Test, Depth of Kanban, Scrum Self Assessment, and Agility Questionnaire. When all surveys are examined, it is seen that almost all of them have at least one fully achieved feature. Therefore, all of

the surveys are helpful in some degree based on the purpose of usage.

As a further research, we are planning to widen the case study to all available agile maturity self-assessment surveys. Although not frequently referenced there are currently a number of surveys such as SAFe Team Self-Assessment [28], Scrum Checklist [29] and Agile Assessment [30] uncovered by this study.

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REFERENCES

- [1] B. Aysolmaz and O. Demirörs, "A Detailed Software Process Improvement Methodology: BG-SPI," *Commun. Comput. Inf. Sci.*, vol. 172, pp. 97–108, 2011.
- [2] A. Uskarci and O. Demirörs, "Do staged maturity models result in organization-wide continuous process improvement? Insight from employees," *Comput. Stand. Interfaces*, vol. 52, no. March 2016, pp. 25–40, 2017.
- [3] S. Jalali, C. Wohlin, and L. Angelis, "Investigating the applicability of Agility assessment surveys: A case study," *J. Syst. Softw.*, vol. 98, pp. 172–190, 2014.
- [4] M. Cohn and K. Rubin, "Comparative Agility," 2008. [Online]. Available: <https://comparativeagility.com/>.
- [5] Thoughtworks, "Agile Assessment," 2010. [Online]. Available: <https://www.agileassessments.com>.
- [6] T. J. Gandomani and M. Z. Nafchi, "Agility assessment model to measure Agility degree of Agile software companies," *Indian J. Sci. Technol.*, vol. 7, no. 7, pp. 955–959, 2014.
- [7] O. E. Adalı, Ö. Ö. Top, and O. Demirörs, "Evaluation Of Agility Assessment Tools: A Multiple Case Study," in *16th International Conference, SPICE*, 2016.
- [8] J. Sutherland, "ScrumButt Test aka the Nokia Test," 2009. [Online]. Available: <http://jeffsutherland.com/ScrumButtTest.pdf>.
- [9] Ö. Ö. Top and O. Demirörs, "A Reference Model for Software Agility Assessment: AgilityMod," in *15th International Conference, SPICE*, 2015, pp. 145–158.
- [10] Ö. Ö. Top and O. Demirörs, "AgilityMOD : Software Agility Assessment Reference Model v3 . 0 Application : Case Study Results," Technical Report METU/II-TR-2014-40. Informatics Institute, Middle East Technical University, 2014.
- [11] Ö. Ö. Top and O. Demirörs, "Assessment of Agile Maturity Models: A Multiple Case Study," in *13th International Conference, SPICE*, 2013.
- [12] C. Patel and M. Ramachandran, "Agile Maturity Model (AMM) : A Software Process Improvement framework for Agile Software Development Practices," *International Journal of Software Engineering*, vol. 2, no. 1, pp. 3–28, 2009.
- [13] R. Benefield, "Seven Dimensions of Agile Maturity in the Global Enterprise: A Case Study," in *System Sciences (HICSS), 2010 43rd Hawaii International Conference*, 2010, pp. 1–7.
- [14] J. Nawrocki, B. Walter, and A. Wojciechowski, "Toward maturity model for eXtreme programming," in *Proceedings 27th EUROMICRO Conference. 2001: A Net Odyssey*, 2001, pp. 233–239.
- [15] A. P. G. Yin, "Scrum Maturity Model," Technical University of Lisbon, 2011.
- [16] J. Packlick, "The Agile Maturity Map A Goal Oriented Approach to Agile Improvement," *Proc. - Agil. 2007*, pp. 266–271, 2007.
- [17] M. Leppanen, "A Comparative Analysis of Agile Maturity Models," in *Information Systems Development: Reflections, Challenges and New Directions*, vol. 1, Springer New York, 2013, pp. 329–343.
- [18] ISO/IEC, "ISO 15504-5 Information Technology - Process Assessment - Part 5: An exemplar Process Assessment Model," 2006.
- [19] K. Waters, "How Agile Are You? (Take This 42 Point Test)," 2008.
- [20] Marberg Consulting, "Karlskrona Test Online," 2008. [Online]. Available: <http://mayberg.se/learning/karlskrona-test-online>.
- [21] M. Britsch, "Assessing your Client's Agility - An Agility Questionnaire," 2014. [Online]. Available: <http://www.thedigitalbusinessanalyst.co.uk/2014/07/Agile-Questionnaire.html>.
- [22] Borland, "Borland Agile Assessment 2009," 2009. [Online]. Available: http://borland.typepad.com/agile_transformation/2009/03/borland-agile-assessment-2009.html.
- [23] C. Achouiantz, "Depth of Kanban - A Good Coaching Tool," 2013. [Online]. Available: <http://leanagileprojects.blogspot.com.tr/2013/03/depth-of-kanban-good-coaching-tool.html>.
- [24] M. James, "An Example Checklist for ScrumMasters," 2007. [Online]. Available: http://scrummasterchecklist.org/pdf/ScrumMaster_Checklist_12_unbranded.pdf.
- [25] I. Cape Project Management, "Scrum Self Assessment," 2015. [Online]. Available: <http://www.agileprojectmanagementtraining.com/Agile-Self-Assessment/quiz.html>.
- [26] ISO/IEC, "ISO 15504-2 Information Technology - Process Assessment - Part 2: Performing an Assessment," 2003.
- [27] CMMI Product Team, "CMMI® for Development, Version 1.3 CMMI-DEV, V1.3," *Software Engineering Institute, Carnegie Mellon University*, no. November, Pittsburgh, Pennsylvania, 2010.
- [28] Scaled Agile, "Team Self-Assessment - Scaled Agile Framework," 2008. [Online]. Available: <http://www.scaledagileframework.com/?wpdmact=process&did=NjEuaG90bGluaw==>.
- [29] Henrik Kniberg, "The Unofficial Scrum Checklist," 2010. [Online]. Available: <https://www.crisp.se/wp-content/uploads/2012/05/Scrum-checklist.pdf>.
- [30] P. Nowinski, "Agile Assessment," 2016. [Online]. Available: <https://nowinskipttr.wordpress.com/2016/04/29/agile-assessment/>.